

Lesson 1 · Homework

NCEA Level 2 · AS 91262 · Gradients, differentiation and tangents

Time: about 60 minutes

Name: _____

Date: _____

Show all working and write $f'(x)$ each time. [A] Achieved, [M] Merit, [E] Excellence.

1. Differentiate.

[A]

(a) $f(x) = 4x^5 - 2x^3 + 9$

.....

(b) $y = 7 - 3x + 5x^2$

.....

(c) $g(x) = \frac{1}{2}x^4 - \frac{1}{3}x^3 + x$

.....

(d) $y = (x + 4)(x - 1)$

.....

.....

2. Differentiate $y = x^2(3x - 5)$.

[A]

.....

.....

3. Differentiate $f(x) = (2x - 3)^2$.

[A]

.....

.....

4. Find the gradient of $f(x) = x^3 + 2x^2 - 5x + 1$ at the point where $x = 2$.

[A]

.....

.....

5. Find the gradient of $y = 4 - 3x^2$ at the point where $x = -2$. [A]

.....

.....

6. For $y = x^4 - 8x$, find the gradient at $x = 1$. Is the curve going up or down there? [A]

.....

.....

7. Find the coordinates of the point on $y = x^2 - 6x + 10$ where the gradient is 4. [A]

.....

.....

8. Find the x -values where $f(x) = x^3 - 3x^2 - 9x + 2$ has gradient 0. [A]

.....

.....

.....

9. Find the coordinates of the two points on $y = x^3 - 12x + 1$ where the gradient is 15. [A]

.....

.....

.....

10. The curve $f(x) = ax^3 + 2x^2 - 1$ has gradient 32 at $x = 2$. Find a . [A]

.....

.....

11. Find the equation of the tangent to $y = 3x^2 - x + 2$ at $x = 1$. [A]

.....

.....

.....

12. Find the equation of the tangent to $f(x) = x^3 - 4x$ at $x = -1$. [A]

.....

.....

.....

13. The curve $y = px^2 + qx$ passes through $(2, 2)$, and its gradient at that point is 5. Find p and q . [M]

.....

.....

.....

14. The tangent to $y = x^2 + 2x - 3$ at the point P is parallel to the line $y = 6x - 1$. Find the coordinates of P and the equation of the tangent. [M]

.....

.....

.....

15. Extension The tangent to $y = x^2$ at the point where $x = a$ passes through $(0, -9)$. Find the possible values of a and the equations of these tangents. [E]

.....

.....

.....

.....

Optional extra: StudyTime Level 2 Calculus Guide, "Stop and Check" on pages 10 and 12.